Where the Galaxy Law Sets the Core: a_0 and the Structure of Cold Dark Energy Halos, from Galaxies to Clusters

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Abstract

The owner's programme studies a galaxy law, $g_N = g^2/(g + a_0)$ with $a_0 = c^2\sqrt{\Lambda/32\pi}$ ($\kappa = \frac{1}{2}$ fitted), and its covariant form v3s. This note follows the record's lanes from that law to what lies beyond it, on both a_0 footings (9.36×10^{-11} and $1.13 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, never pooled).

1. **The law** passes the SPARC radial-acceleration relation (rms 0.054 dex), MeerKAT's a_0, isolated groups, tidal dwarfs, the screened Solar System and binary pulsars.
2. **Beyond galaxies** it fails. Cluster cores sit above it at Z 4.1–5.5. Faint dwarfs carry a real, mass-driven extra. The early universe needs a cold component that is already dust-like by $z \approx 5 \times 10^4$. We call that component *cold dark energy*: a gravitating $w \approx 0$ field component paired with dark energy, not matter and not a particle.
3. **The doors that close.** Selectors, late or vacuum heating, feedback, early dark energy, recombination release and orbit-keyed conversion all fail, each with a named mechanism.
4. **Route E.** Cold dark energy is everywhere, and the law is the form it takes around baryons. The external-field test in the record does not distinguish a fundamental law from an emergent one.
5. **The core result.** Burkert cores with $\rho_0 r_0 = a_0/(2\pi G)$ (106.9 / 129.2 $M_\odot\,\mathrm{pc^{-2}}$, no new constant) pass SPARC and the cluster outskirts, but under-predict the heavy cluster cores. The rule $\Sigma_{\rm core} = \max(a_0/2\pi G,\ \eta\,\Sigma_{\rm coll})$, with $\eta = 14.83$ from self-similar infall and an *external* formation concentration $c_f = 4$, passes SPARC and 14 of 15 cluster keys, including the heavy cores. It fails two keys narrowly: groups ($Z = +2.05$) and the gas-richest clusters' outskirts ($Z = +2.41$).
6. **Deriving c_f.** Scale-free infall gives 1.85 and fails; spherical collapse of the initial peak profile also gives 1.85 and fails (post hoc, exploratory). Collisionless zoom simulations in the record give a native c_f of 7.2 / 7.0 / 5.6 at 10^{12} / 10^{13} / $10^{14}\,M_\odot$, above the data's window and not converged (uncertain by about a factor 2). With that native c_f the rule passes SPARC, groups and every cluster key but one, the 326-cluster inner band, which it over-predicts ($Z = -2.41$). This is the closest result in the record with no borrowed number; more seeds are running.
7. **Dwarfs.** The same a_0-floor halos, with nothing new fitted, reproduce four of five Local Group dwarf classes and fail the Milky Way ultra-faints ($Z = +3.80$ / $+3.66$, under-predicted).
8. **The tether.** A local vacuum-set floor ($g_T = 0.254\,a_0$) fails growth and self-consistency. A floor keyed on bound objects is consistent but nonlocal, and has no action.
9. **Current picture.** a_0 sets the cold dark energy core floor from dwarfs to large spirals; above galaxy scale, collapse plus a native concentration sets the core. Three items are open: the cluster inner band with the convergence of c_f, the ultra-faint halo masses, and the tether mechanism.
10. **A map, for orientation only.** A Cosmicflows-4 flow map with the SPARC galaxies and nearby eRASS1 clusters on it illustrates the picture. It is not evidence for the theory.

1 Setting and rules

- $\kappa = \frac{1}{2}$ is fitted. Footings: canonical $a_0 = 9.3603 \times 10^{-11}$, alternative $1.1312 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, never pooled; pairs below are canonical / alternative. Each lane froze its criteria before computing, ran a mutation control that must fail, and was rerun by the coordinator; the record's summary is `V3S_FINAL_VERDICT_2026-10-24.md` (§§7–8) and `COLD_SECTOR_SPEC_2026-10-24.md`. Lane folders are in `rebuild_2026_10_22/`.
- **Naming.** *Cold dark energy* is the cold, clumping ($w \approx 0$) component that the data require beyond the law. It is a gravitating field component paired with dark energy ($w \approx -1$). It is not matter and not a particle; "dust-like" describes only how it clusters. ΛCDM is the name of the external rival model.
- No result here is evidence for the framework over ΛCDM or any other rival. Failures are stated with their mechanisms.

2 The galaxy law

In the static limit the law is AQUAL with $\mu(x) = x/(1+x)$, i.e. $g_N = g^2/(g+a_0)$, and $a_0 = \kappa c\sqrt{G\rho_\Lambda} = c^2\sqrt{\Lambda/32\pi}$. Its covariant form, v3s, is a Khronon action with a switched stiffness term (PAPER60; the verdict, §1). Its constraint algebra closes with 3 degrees of freedom, and the switched stiffness is well posed with the statics unchanged (qualifiers in the record). Table 1 lists what the law passes.

Table 1: What the galaxy law passes (canonical / alternative).

test	result	lane
SPARC radial acceleration	rms 0.054 / 0.054 dex, one Υ per galaxy (median 0.52 / 0.44); the χ^2 line fails, so conditional	P4
MeerKAT a_0	+0.032 dex from SPARC's, same recipe	P4
isolated groups	+0.010 / −0.010 dex from the law on baryons	P4
tidal dwarfs	$+0.055 \pm 0.114$ dex: Newtonian, as predicted	U1
Solar System (screened)	all 24 Cassini cells pass, by 4.8×10^7 to 5.1×10^9; the unscreened law fails	C6
binary pulsars	at most 3.2×10^{-14} of the double pulsar's precision	C8

As a theory of gravity and cosmology, v3s fails: clusters, growth and the CMB, the radiation era, hosted dwarfs and the Milky Way disc (PAPER60; verdict §3). Most of these follow from one mechanism: v3s has one cold field, and kinematics fix it to be the field that carries the law. The rest of this note asks what lies beyond the law.

3 What lies beyond galaxies

- **Cluster cores.** In 0.3–1 Mpc the cores sit 1.5–1.7 times above the law on their baryons. The statistical significance is Z 4.5–5.5 (X11); after X13's correction for a shear-dependent source-size selection it is Z 4.1–5.0, still real by X7's rule. Known baryons fall short by a factor 4–7 (X21).
- **Small dwarfs: a real, mass-driven floor.** Against the isolated law, hosted dwarfs carry an extra that is not tidal and not binaries. The tidally safe subset ($\tau \leq 0.1$) still sits +0.48 / +0.45 dex high, Z = 5.8 / 5.4 (`X20_hosted_dwarf_disequilibrium`). Being hosted adds nothing: field dwarfs of the same stellar mass carry the same extra. A velocity floor $\sigma^2 = \sigma_{\rm iso}^2 + c_0^2$, with $c_0 = 4.6$ / 4.1 km s^{-1} calibrated on field dwarfs only, cuts the hosted χ^2 from 637 to 217 (canonical) and 559 to 195 (alternative) (X24). Tides at pericentre are ruled out: dwarfs with weak pericentric stress still sit +0.60 / +0.55 dex high, $Z = 4.8$ / 4.4 (X25). The floor is not in the action (`X25_floor_from_action`).
- **The early universe.** The Lyman-α forest needs dust-like clustering down to $\lambda_{\rm hm} \approx 0.42\,{\rm Mpc}/h$ ($M_{\rm hm} \approx 4.9 \times 10^9\,M_\odot$). Gravity alone then drops the component into galaxies: median capture 0.76 of the fair share (X22). In patched CLASS, a cold component switched on at recombination is excluded (σ_8 at 0.050 / 0.047 of ΛCDM's for a turn-on at $z = 1100$ / 1500). It must already be dust-like by $z \approx 5 \times 10^4$ (X34).
- **Its nature.** It must be collisionless and concentrated toward cluster centres. Any component added on top of the law, retained hydrostatically or partially, spreads mass outward: its predicted cluster OUTER band is 1.42–2.26 against measured 0.72 ± 0.14 / 0.66 ± 0.13 (X30). In the Bullet Cluster a dust-like tracer with dispersion near the well's survives ($f_{\rm ret}$ 0.80–1.02) and lands on the galaxies (X32).

In every test run in the record, the cold dark energy the data require behaves like the cold component of ΛCDM. No test here distinguishes it from that component. What the framework adds is the galaxy law, and with it the question of why isolated galaxies show no extra today.

4 The doors that close

Table 2 lists the mechanisms tried for keeping cold dark energy where the data want it, and how each fails.

The general statement (X37) is that any depletion rule keyed to orbits, density, temperature or time removes cold dark energy from cluster centres first, where the data want it most.

Table 2: Doors tried for the cold sector (both footings unless noted).

door	how it fails	lane
one selector variable	no single catalogue variable or axis-aligned pair separates the classes that need extra from those that need none	X19
feedback energy	$F = e_{\rm fb}/\sigma^2$ does not order the classes (Spearman +0.10); the heavy clusters that need extra have the lowest F	X31
universal late heating	clusters capture the whole fair share: $A_{\rm MID}$ 7.5–10.7 against 1.0–2.2	X23
late heating, graded	hosted dwarfs get $\leq 8 \times 10^{-4}$ dex against +0.21 to +0.64; clusters T2/T3 are over-captured	X23
vacuum heating	no ϵ window: the forest allows $\epsilon \leq 2.41 \times 10^{-6}$, SPARC needs $\geq 7.65 \times 10^{-6}$ / 8.59×10^{-6}	X24b
a heated sheet	survives the cluster terciles at 45 / 37 grid points, then all are excluded by the outskirts profile	X26, X30
early dark energy	UNDECIDED; v3s has nothing for a large early a_0 to act on ($\rho_\tau \equiv 0$)	X35
recombination release	excluded at recombination; reduces to ordinary dust-like clumping by $z \approx 5 \times 10^4$	X34
orbit-keyed conversion	empties the cores first: OUTER 1.67 against 0.72 ± 0.14 ($Z = -6.7$); MID slope with KT of the wrong sign	X37

5 Route E: the law as the form cold dark energy takes

Route E puts cold dark energy everywhere, keeps gravity Newtonian, and reads the galaxy law as the form cold dark energy halos take around baryons. It inherits growth, the forest, the Bullet Cluster and the tidal dwarfs by construction.

- **With cusps it fails softly on SPARC (EM1).** NFW halos on one global SPARC halo-mass relation give rms 0.0656 but median Υ 0.36, below the stellar-population window. With Υ pinned at 0.5 for both, the dust model gives 0.1020 against the law's 0.1124 / 0.1029. SPARC alone does not separate them.
- **The external field does not decide it (X36).** Large SPARC galaxies' residuals fall with the external field: slope -0.561 ± 0.175 / -0.525 ± 0.176. The law's own external-field effect predicts almost none of it ($Z = -3.21$ / -2.99). Route E reaches part of it ($Z = -2.26$ / -2.06). The two predictions differ by only 0.71 / 0.70 of the measurement's error, so the lane is NOT DIAGNOSTIC. Gaia DR4 wide binaries cannot separate them either: both readings predict Newtonian binaries.

Stated plainly: no test in the record yet distinguishes a fundamental law from a law that emerges from cold dark energy around baryons. In route E with cusps, the link $a_0 \propto \sqrt{\Lambda}$ would be a coincidence. Sections 6 and 7 give a_0 a role inside route E: it sets the cold dark energy core floor from dwarfs to large spirals.

6 The core surface density

X38: cores set by a_0. Burkert-cored cold dark energy halos with $\rho_0 r_0 = a_0/(2\pi G)$, i.e. 106.9 / 129.2 $M_\odot\,{\rm pc}^{-2}$, add no new constant; galaxy halo masses use EM1's committed relation, and cluster halos take the fair share. SPARC passes at a sensible Υ: rms 0.0795 / 0.0729, median Υ 0.77 / 0.60. The cluster OUTER band passes for the first time in the record (all-KT $Z = +0.14$ / -0.06), where every added-component model overshot. The heavy cores fail, under-predicted: MID T3 $Z = +4.26$ / +3.80, MID slope with KT +3.07 / +2.99, and the heaviest gas tercile's OUTER band (MGAS T3) +5.36 / +4.94. Its cluster cores come out too wide (0.44 Mpc in the hottest tercile), and it also fails the groups line ($Z = +2.48$ / +2.29, X40b).

What does not raise the cluster cores.

- **Contraction (X39).** Blumenthal contraction by the BCG and hot gas moves MID T3 only from 1.05 to 1.13 against 2.13, and it rebuilds galaxy cusps: SPARC median Υ falls to 0.39 / 0.32 and fails. Any process that densifies cluster cores through baryons densifies galaxy cores first.
- **A fixed core radius (X40).** The best SPARC value, $r_c = 9.73$ kpc, gives rms 0.119 at Υ 1.20 (fails), and overshoots the clusters.
- **Framework-native rules (X40b).** Setting the core by $\max(a_0, g_b)$ never switches on: at cluster cores g_b is only 0.04–0.07 a_0. Setting it by a_0 at the formation epoch has the right sign but reaches at most a factor 1.34.

With galaxies pinned, the clusters need $S \propto M^{\alpha}$ with $\alpha_{\rm best} = 0.30$ / 0.25, i.e. cores at 3–5 times $a_0/(2\pi G)$.

Collapse supplies the slope (X41). The Lacey–Cole formation epoch on the record's own $P(k)$ gives the collapsed object's ρr a slope of 0.23 in mass over 10^{13}–$10^{15}\,M_\odot$, close to the needed 0.25–0.30. Its amplitude is only 0.26 / 0.22 of $a_0/(2\pi G)$ at the cluster median, so a core-concentration factor η of 11.4–19.0 / 13.8–23.0 is needed. X42 calibrated a self-similar rise on the cluster cores alone: it implies a factor of 9.0–14.4 over the same collapse ρr, and it fails the groups out of sample ($Z = +2.41$ / $+2.27$).

X43: floor plus collapse core.

$$\Sigma_{\rm core}(M) = \max\Big(\frac{a_0}{2\pi G},\ \eta\,\Sigma_{\rm coll}(M)\Big)\,, \qquad \eta = \frac{4}{3B(1)}\,c_f^{5/4} = 14.83 \quad [c_f = 4 \text{ EXTERNAL}].$$

Cold dark energy collapses like collisionless dust (Bertschinger's self-similar infall, $\rho \propto r^{-9/4}$); its core is the inner $1/c_f$ of the half-mass object at the Lacey–Cole formation epoch; and it can never be cored more thinly than the floor $a_0/(2\pi G)$. Nothing is fitted; $c_f = 4$ is an external value from collapse studies. The crossover falls at $M \approx 10^{12.1}$ / $10^{12.4}\,M_\odot$: galaxies sit on the a_0 floor, groups and clusters on the collapse core. Table 3 shows the grades.

Table 3: Core rules graded on the same keys (canonical; alternative in brackets where it differs in class). Measured MID T3 2.13, MGAS T3 OUTER 1.37. X44 sets c_f from self-similar infall timing; X45 (peak collapse) gives the same 1.85 and the same grades. X48 uses the native $c_f(M)$ measured at ng 48 (7.2 / 7.0 / 5.6, not converged).

key	X38 (floor only)	X43 ($c_f = 4$, external)	X43, $c_f = 5$ (reported)	X44 ($c_f = 1.85$)	X48 (native, ng 48)
SPARC rms / median Υ	0.0795 / 0.77	0.0782 / 0.77	0.0721 / 0.72	pass	0.0650 / 0.64
groups Z	+2.48	+2.05	+1.82	+2.48 (+2.29)	+1.53
MID T3	1.05, $Z = +4.26$	2.06, $Z = +0.30$	—	1.40, $Z = +2.90$	2.34, $Z = -0.82$
MID slope with KT, Z	+3.07	+1.70	—	+2.49	+1.32
MGAS T3 OUTER	0.81, $Z = +5.36$	1.12, $Z = +2.41$	fails	0.95, $Z = +4.07$	1.18, $Z = +1.80$
INNER band (all clusters)	pass	pass	—	—	4.81, $Z = -2.41$
cluster keys passing	10 / 15 (11)	14 / 15	14 / 15	—	14 / 15

What X43 passes and fails. SPARC passes (rms 0.0782 / 0.0729, median Υ 0.77 / 0.60). The heavy cores pass: MID T3 moves from $Z = +4.26$ to +0.30 and the all-MGAS OUTER band from +2.36 to +0.49. Two keys fail, both slightly under-predicted: groups at $Z = +2.05$ on both footings (line 2.0), and the MGAS T3 OUTER band, 1.12 against 1.37 ($Z = +2.41$). Reported only, not adopted: $c_f = 5$ ($\eta = 19.6$) passes the groups ($Z = +1.82$).

The status of c_f. It is external, and two keys fail; X43 is not a pass. A first derivation from scale-free self-similar infall timing (X44) gives $c_f = 2^{8/9} = 1.85$ ($\eta = 5.66$), which fails four keys: groups, MID T3, the MID slope and MGAS T3 OUTER. The data pin $c_f \approx 4$–4.2: X43's groups line crosses $Z = 2.0$ at $c_f \approx 4.21$ / 4.20, and X42's cluster-only calibration gives 3.3–4.6. That is the standard formation concentration. It needs the structure of the initial density peaks and cannot come from scale-free infall; the framework does not derive it.

Peak collapse does not give c_f either (X45, post hoc, exploratory). Spherical collapse, shell by shell, of the mean (BBKS) profile around a Gaussian peak on the record's own $P(k)$, with the half-mass object at the Lacey–Cole formation epoch, gives no -2 log slope at any mass from 10^{10} to $10^{15.5}\,M_\odot$. The slope is about 0 over 75 % of the shells and only -0.6 at the edge: all shells collapse at nearly the same time, so the formation object is a near-uniform core. Every c_f falls back to 1.85, and the regrade equals X44 (groups $Z = +2.48$ / $+2.29$, MGAS T3 OUTER $Z = +4.08$): **PK FAILS** in every run on both footings. Two amendments came after the first result, so this verdict is post hoc and exploratory. Reading: $c_f \approx 4$ comes from shell crossing, orbit mixing and relaxation, which no spherical shell-by-shell model captures.

A native c_f from the record's own simulations (X48). Collisionless zooms of constrained peaks (cold dark energy plus Λ on the record's background, no a_0 floor; not v3s) measure c_f from the -2 crossing at three masses and three seeds. Two stages, both stated:

- **ng 38: UNRESOLVED.** Medians 10.6 / 5.9 / 11.0 at 10^{12} / 10^{13} / $10^{14}\,M_\odot$, but the frozen resolution check fails: at 10^{14}, seed 1, ng 38 → 48 moves c_f from 12.68 to 2.59. The ng 38 numbers are not evidence.
- **ng 48 (amendment A1): NATIVE c_f ABOVE WINDOW**, at ng 48; convergence beyond ng 48 is not tested. Medians 7.2 / 7.0 / 5.6, against the window [3.3, 4.6]. Seed scatter is a factor 4–6 (2.3–13.4) and the per-run ng 38 → 48 shifts run from −80 % to +48 %, so the medians are uncertain by about a factor 2.
- **X43 regraded with the native $c_f(M)$** (η 30.7 / 29.7 / 22.7 against 14.83) passes SPARC (rms 0.0650 at Υ 0.64 / 0.0657 at 0.53), the groups ($Z = +1.53$), every MID key with its slope, every OUTER key, and MGAS

T3 OUTER ($Z = +1.80$). It fails one key on both footings: the inner band of all 326 clusters, G-IN-Mall, over-predicted (4.81 against 2.28 ± 1.05, $Z = -2.41$; -2.06 in the $h = 0.686$ cell set).

- **Status.** This is the closest result in the record with no borrowed number, and it is not converged. Amendment A2 adds seeds 4–9 at ng 48 (18 runs); they were running when this draft was built, and no A2 number is used here. No single c_f passes all 15 cluster keys: at $c_f = 4$ the inner band passes and the groups and MGAS T3 OUTER fail; at the native values the reverse.

Reading. The cold dark energy core surface density is $a_0/(2\pi G)$ in galaxies. Above galaxy scale it is the collapse value raised by a formation concentration, which the record's own simulations now measure, though not yet to convergence. In this picture $a_0 = c^2\sqrt{\Lambda/32\pi}$ is a constant of the cold sector, and the galaxy law is what the a_0 floor looks like. κ is still fitted, and adopting the picture is the owner's decision, not a lane result.

7 The floor below galaxies, and what holds it

Dwarfs on the a_0 floor (X49). Every Local Group dwarf sits far below the crossover ($\Sigma_{\rm coll} \leq 3.8 \ll 107\,M_\odot\,{\rm pc}^{-2}$), so under X43's rule its halo is a pure a_0-floor Burkert halo, independent of c_f. Halo masses come from EM1's SPARC relation, extrapolated downward; nothing new is fitted. Table 4 gives the per-class median offsets E (dex) of the measured dispersions.

Table 4: X49: dwarf classes on pure a_0-floor halos (median offset E in dex, with Z).

class (N)	canonical E (Z)	alternative E (Z)	median M_{200} ($M_\odot$, extrapolated)	class
LG field (13)	-0.034 (-0.49)	-0.068 (-0.95)	2.7×10^8	reproduced
MW classical (8)	$+0.125$ ($+0.91$)	$+0.081$ ($+0.56$)	4.0×10^7	reproduced
bright M31 (25)	$+0.176$ ($+1.70$)	$+0.161$ ($+1.55$)	2.9×10^7	reproduced
M31 others (5)	$+0.571$ ($+1.71$)	$+0.544$ ($+1.59$)	3.1×10^6	reproduced
MW ultra-faint (31)	$+0.476$ ($+3.80$)	$+0.459$ ($+3.66$)	5.2×10^5	UNDER

FAIL: MW ultra-faint (UNDER) on both footings; the other four classes are reproduced with no new constant. The ultra-faints want heavier halos than the extrapolation gives (the extrapolation runs three to five decades below SPARC's range, and is the disclosed weak link); the fair-share variant, with even smaller halos, is worse ($Z = +8.3$). The dwarf mass trend survives (Spearman -0.64 / -0.65 with $\log M_\star$). Tidal stripping is not modelled; it would worsen the ultra-faints. **Consequence:** the a_0 floor replaces v3s's separate dwarf velocity floor c_0 for every class except the ultra-faints.

The tether: a vacuum-set floor (X47, the owner's question). What keeps cold dark energy at the floor? Hypothesis VT: it stays bound only where its own pull exceeds g_T, fixed (not fitted) as the peak self-pull of an $a_0/(2\pi G)$ Burkert core, $g_T = 0.254\,a_0$ (2.38×10^{-11} / $2.88 \times 10^{-11}\,{\rm m\,s^{-2}}$). For a fixed profile shape, "peak pull $= g_T$" is the same statement as "Σ = constant", so the floor itself restates X38; the content is in where the key acts.

- **As a local rule (VT-L): FAILS** at growth and self-consistency, on both footings. The linear field's own pull today is at most $0.020\,g_T$, so a local key tethers the linear field and removes its late growth (the μ-pincer in acceleration form). A latching key never forms a floor. And an at-floor cored halo is itself below g_T at every radius but one, at its centre and in its envelope.
- **As a rule about bound objects (VT-O): CONSISTENT, but nonlocal**, and no action is written for it. Collapse itself is below threshold, so the key must act after virialisation.
- **Data.** Per-galaxy free-Σ fits on SPARC centre on g_T (median $+0.07$ / -0.02 dex, Υ free) with a spread of ± 0.32 dex: centred, but no sharp pile-up. Only 10–13 % of galaxies sit more than 0.3 dex below the floor. As context only (PROVISIONAL transcription), Gentile et al. (2009) give 1.32 / $1.10\,g_T$.
- **Prediction.** If a_0 tracks dark energy, $\Sigma_{\rm floor}(z) \propto a_0(z)$ shifts by only -0.06 dex at $z = 2$. The record's $z \approx 2$ samples hold tens of galaxies and cannot test that as they stand.
- **Reading.** Whatever tethers cold dark energy must key on bound objects (halos), not on any local acceleration, density or expansion invariant. The mechanism is open.

The current picture. a_0 sets the cold dark energy core floor, $\rho_0 r_0 = a_0/(2\pi G)$, from dwarfs to large spirals (X38, X49). Above galaxy scale, gravitational collapse plus a native formation concentration sets the core (X43, X48). Three items are open: the cluster inner band together with the convergence of the native c_f; the halo masses of

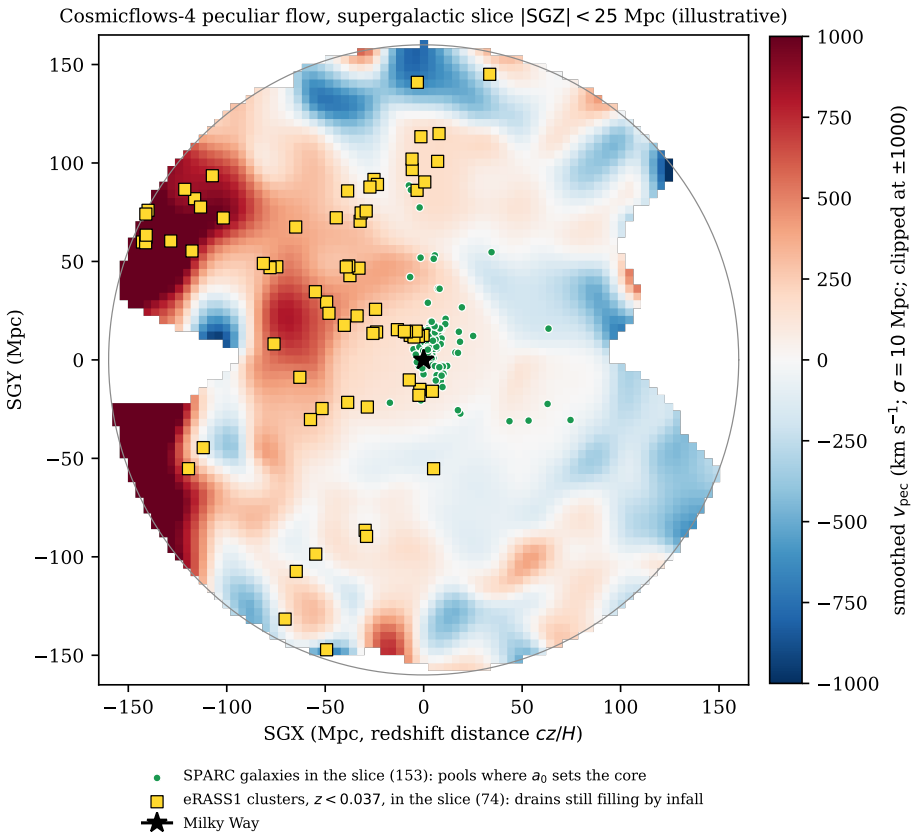


Figure 1: Measured Cosmicflows-4 peculiar flow in the supergalactic slice $|\mathrm{SGZ}| < 25\,\mathrm{Mpc}$, $R < 160\,\mathrm{Mpc}$ (MAP1). Colour: smoothed peculiar velocity ($\sigma = 10\,\mathrm{Mpc}$ kernel, inverse-variance weights, shown where at least 3 effective groups contribute), clipped at $\pm 1000\,\mathrm{km\,s^{-1}}$; red recedes faster than the Hubble flow. Green dots: SPARC galaxies in the slice, labelled "pools where a_0 sets the core". Yellow squares: eRASS1 clusters at $z < 0.037$ in the slice, labelled "drains still filling by infall". The labels are the reading of Section 6, not measurements. Illustrative only: the map is not evidence for the theory.

the ultra-faints; and the tether mechanism, which must be object-keyed.

8 The cosmic river map (illustrative)

Figure 1 places the two regimes of Section 6 on the measured local flow. MAP1 groups the 38,035 Cosmicflows-4 galaxy groups, self-calibrates $H = 73.56\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ on the catalogue's own distance scale, and computes peculiar velocities with the Watkins–Feldman estimator. The slice holds 5,875 groups, 153 SPARC galaxies (at X33's distances) and 74 of the 178 eRASS1 clusters within 160 Mpc. No fit is made and no verdict is drawn.

Caveats. Per-group peculiar velocities are noisy (median error $905\,\mathrm{km\,s^{-1}}$). No inhomogeneous-Malmquist correction is applied, so the field at large R is biased. eRASS1 covers only the western Galactic hemisphere, so the cluster positions trace the survey footprint as much as the cosmic web. The map shows where the two regimes sit; it tests nothing.

9 Discussion

The current picture. a_0 sets the cold dark energy core floor from dwarfs to large spirals; above galaxy scale, collapse plus a native formation concentration sets the core (Sections 6 and 7).

Open.

- **The cluster inner band and the convergence of c_f.** With $c_f = 4$ (external) the inner band passes and two keys fail narrowly (groups, MGAS T3 OUTER). With the native c_f measured at ng 48 those two pass and the inner band of all 326 clusters is over-predicted ($Z = -2.41$). The native medians are not converged (uncertain by about a factor 2), and amendment A2's further seeds are running. Analytic routes do not give c_f: scale-free infall (X44) and spherical peak collapse (X45) both give 1.85. The MGAS T3 OUTER band also failed X40b's power law, X42's two rules and X43, and the verdict asks for a data-side check of that band before more rules are built on it.
- **The ultra-faint halo masses.** The a_0 floor reproduces four dwarf classes, but the MW ultra-faints want heavier halos than EM1's relation gives when extrapolated far below its range (X49). A framework-native halo-mass relation below $10^8\,M_\odot$ of baryons would be needed.
- **The tether mechanism.** A local vacuum-set floor fails; whatever holds cold dark energy at the floor must key on bound objects, and no action for such a key exists in the record (X47).
- **Fundamental or emergent?** No test in the record separates a fundamental law from a law that emerges from a_0-cored cold dark energy (X36, EM1).
- **What lies outside the picture.** Nothing here is a cosmology, and κ is still fitted.

Data still to come (dated, sealed before the data): the Khronon-native L1 512^3 growth run (in the compute queue; the converged ladder says the nonlinear hold term does not restore growth); ALMA U4 (2026-11-06) and GS4 (2026-11-28); Gaia DR4 wide binaries and the Milky Way (2026-12-02).

10 What this is not

- Not closed: κ is fitted; $c_f = 4$ is external, and the native c_f is not converged; the groups and MGAS T3 OUTER keys fail at $c_f = 4$, the inner band fails at the native c_f; the ultra-faints fail; the tether has no action.
- Not evidence for the framework over ΛCDM or any other rival. The cold dark energy required here behaves, in every test run, like the cold component of ΛCDM.
- No particle species is added: cold dark energy is a gravitating field component.
- The river map is illustrative, not a test.

Disclosures. X43's criteria were frozen at 00bc1a7950 with no amendment; one loader path fix preceded the first graded run. X44's result (66273956b2) was committed while this note was drafted; it is folded into the verdict (58a2d738f8, coordinator rerun identical apart from timing) and into `COLD_SECTOR_SPEC_2026-10-24.md`. Its 4.21 / 4.20 crossing is in its `.out`; the 3.3–4.6 range is in its README and the verdict, not its JSON. X38's groups failure was found in X40b, not X38. X42's H1 has one cluster-calibrated amplitude, so its SPARC pass is by construction. X24 is post hoc and disclosed in its lane. The verdict document's "107 / 129" is $a_0/(2\pi G)$ rounded; the JSON gives 106.9 / 129.2. Draft 2 adds X45, X47, X48 and X49 from the verdict's §8 (folds f06a515257, dd767f67fa, 7eced79621, ffb12cb0b2, 44907a3ecb). X45 is post hoc and exploratory (amendments A1 and A2 came after the first result; A3, aa70e22e9d, is an implementation fix). X48's simulations were not rerun by the coordinator; its regrade was. Two source discrepancies are resolved here from the JSON: X48's README and result commit quote MGAS T3 OUTER as $Z = -0.24$, which is the hottest tercile's OUTER key (G-OUT-T3); the MGAS T3 OUTER key (G-OUT-M3) is +1.80, as in the verdict. The verdict's inner-band "−2.41 / −2.06" are the $h = 0.672$ and $h = 0.686$ cell sets, not the two footings; both footings give −2.41 in the main set. X47's Gentile et al. comparison is a PROVISIONAL transcription in the record. Two references (Watkins & Feldman 2015; Lacey & Cole 1993) are cited by name from the lanes and are not in the record's bibliography.

11 Data and code availability

All lanes are in `rebuild_2026_10_22/`; commits are criteria / result. Galaxy law: P4 (6c5e6161b9 / 6c0249a8bc), with U1, C6, C8 as in PAPER60. Beyond galaxies: X11 (9a105431c9 / c1cd5455a5), X13 (c4e5a5a064 / 4eeeceba07), X21 (a48f6d5e08 / 3aa4f07b58), `X20_hosted_dwarf_disequilibrium` (87b6832e84 / bd736c128e), X24 (7a79903c3f / 19d31a99d9), X25 (d4827bece5 /

5a754eb2e2), X22 (a2b532e3c6 / 5c7114f508), X34 (6c5d581400 / ab4a3a8e1a), X30 (c6be83f91c / 0f19b6759b), X32 (57ff3e7df4 / 975ccfbf49). Doors: X19 (8960ea2285 / a9ec701f5b), X31 (1aa11f9bcb / 358e3859ed), `X23_late_heated_cold` (b8d81770bb / df896db507), `X23_late_heating_graded` (6ec6b89807 / 9c54fae62e), X24b (1d0f8c6967 / 3492bcbeb4), X26 (b10263968f / c9112e3126), X35 (02848d0330 / 0eb84a459a), X37 (204001f8ad / 538dc083c9). Route E: EM1 (1b0dbd6eef / 889d65276e), X36 (d8651cafc3 / 8d7df6a74d). Core: X38 (9cc80516df / 3e28ca3027), X39 (b75acfc767 / 54684fb62f), X40 (b37983a0f7 / 5dc30a7c1a), X40b (b09db13390 / ba368d2ba1), X41 (0ea05957ee / e9c1b9abe4), X42 (4a13086d02 / 0d7d0d5686), X43 (00bc1a7950 / bf8cc9e7cf), X44 (c8dd02f5aa / 66273956b2), X45 (1fd8e1238a / 8cb58f4d28), X48 (1932858ae8 / aaf492e8b7; A1 2bfdeb83a9 / acb2bf5277 / 737193f3fa; A2 ab49eb27c1). Floor: X47 (bafcb4754e / 9feeb722fc), X49 (fa308d2c47 / 6e93b02215); IC1 (the inner band). Sources read for draft 2: each lane's `README.md` and results JSON, namely `x45_peak_cf_results.json` and `x45_regrade_cfpk_results.json`; `x47_vt_floor_results.json`; `x48_analyze_results.json`, `x48_analyze_results_ng48.json`, `x48_regrade_ng48.json` and `x48_regrade_x43copy_ng48_results.json`; `x49_dwarfs_results.json`. Map: MAP1 (3f9b3ae361). Summaries: `covariant_push_2026_10_22/V3S_FINAL_VERDICT_2026-10-24.md`, `COLD_SECTOR_SPEC_2026-10-24.md`. This note: `PAPER62_fig_river.py` (the figure, from MAP1's JSON) and `PAPER62_core_audit.py` (every number checked against its lane's JSON, `.out` or README; a mutation mode that must fail). Companion: PAPER60 (doi:10.5281/zenodo.23601528). Data: F. Lelli, S. S. McGaugh, J. M. Schombert, AJ 152, 157 (2016) (SPARC); G. Gentile et al., Nature 461, 627 (2009) (context only); R. B. Tully et al., ApJ 944, 94 (2023) (Cosmicflows-4); E. Bulbul et al., A&A 685, A106 (2024) (eRASS1). Theory: E. Bertschinger, ApJS 58, 39 (1985); J. M. Bardeen, J. R. Bond, N. Kaiser, A. S. Szalay, ApJ 304, 15 (1986); J. A. Fillmore, P. Goldreich, ApJ 281, 1 (1984); A. Burkert, ApJ 447, L25 (1995); C. Lacey, S. Cole (1993); R. Watkins, H. A. Feldman (2015). AI assistance: Anthropic's Claude (lanes, review, drafting).